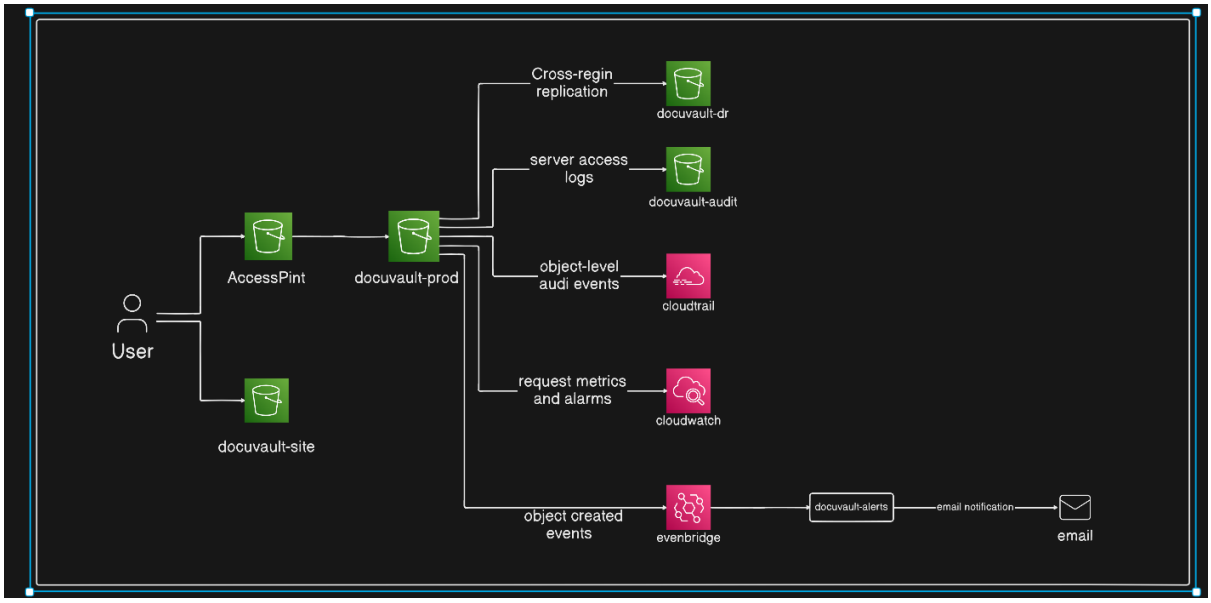
Production-Grade AWS S3 Project: Secure Document Storage and Disaster Recovery

Project name: DocuVault — Enterprise Document Storage Platform

Student name: Abdulslam

1: architecture diagram showing the primary bucket, DR bucket, website bucket, logging, and event destinations



2. S3 Bucket Inventory

The DocuVault project uses multiple Amazon S3 buckets for secure document storage, disaster recovery, website hosting, audit logging, and testing. Each bucket serves a specific purpose and is configured according to its security and storage requirements.

Bucket	Region	Versioning	Encryption	Public access
docuvault-prod	Mumbai (ap-south-1)	Enabled	SSE-KMS	Blocked

docuvault-dr-011430489973-ap-south-2-an	Hyderabad (ap-south-2)	Enabled	SSE-S3	Blocked
docuvault-site	Mumbai (ap-south-1)	Disabled	SSE-S3	Block all public access Off
docuvault-audit-011430489973-ap-south-1-an	Mumbai (ap-south-1)	Disabled	SSE-S3	Blocked
docuvault-lock-test	Mumbai (ap-south-1)	Enabled	SSE-S3	Blocked
docuvault-requester-pays	Mumbai (ap-south-1)	Disabled	SSE-S3	Blocked

Figure 1: S3 bucket list

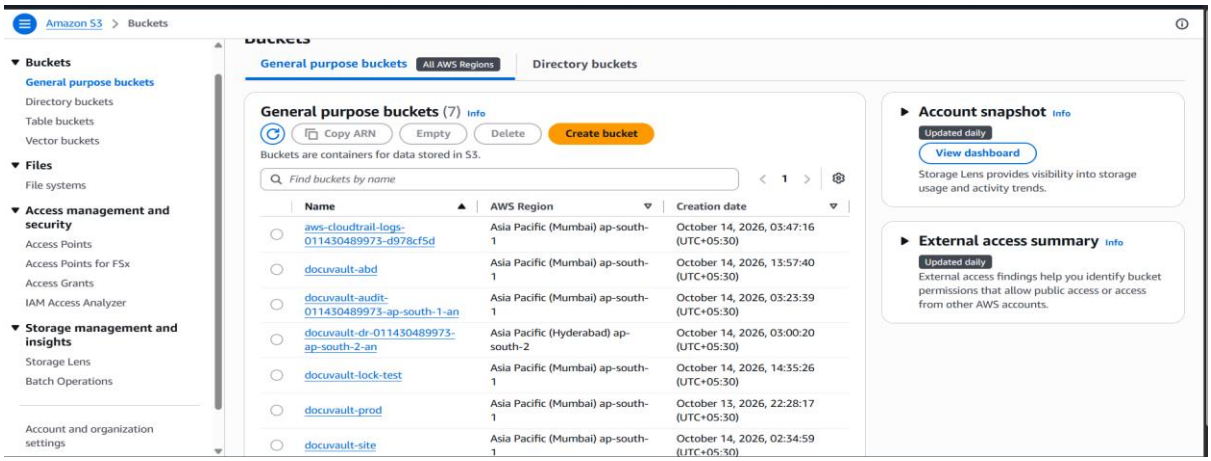


Figure 2: Bucket properties

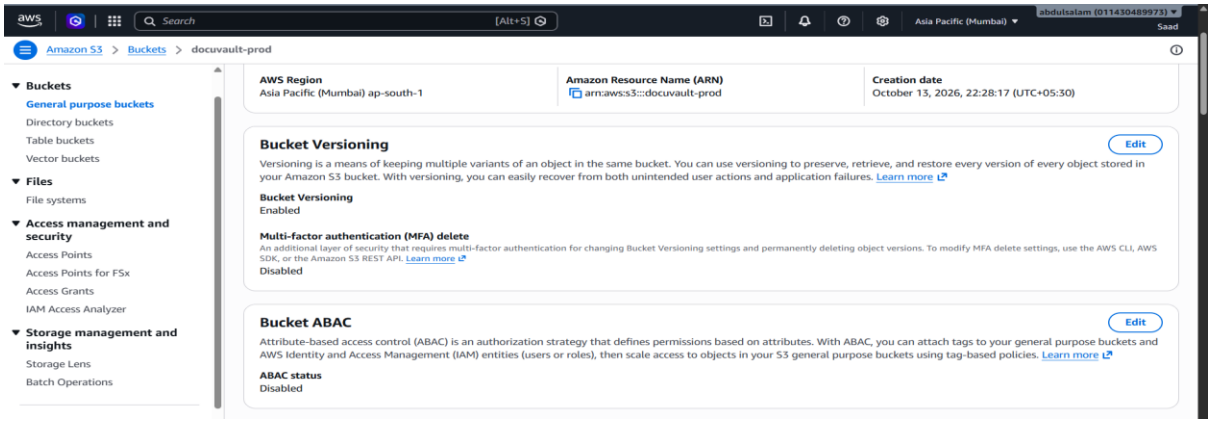
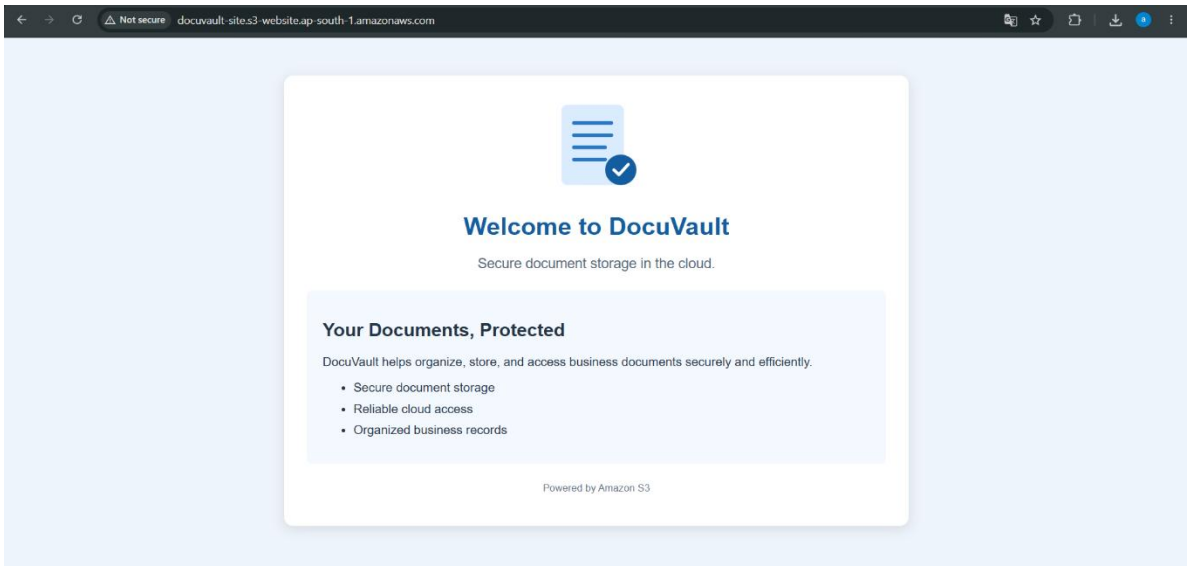


Figure 3 — DocuVault Static Website Hosting Verification



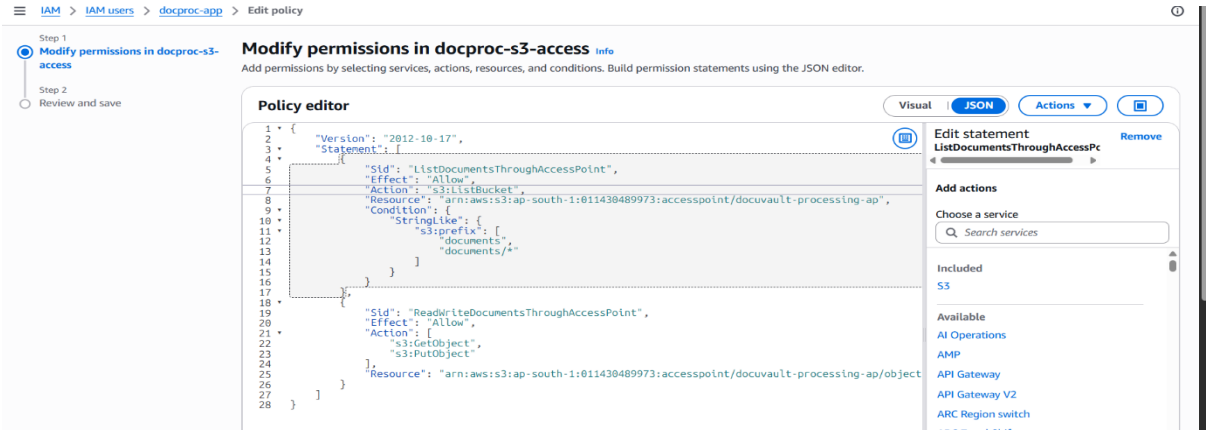
3. IAM Policies and Bucket Access Control

DocuVault uses AWS Identity and Access Management (IAM) and Amazon S3 bucket policies to control access to stored documents. The principle of least privilege is applied so that an application receives only the permissions required to perform its tasks.

3.1 IAM Policy — Least-Privilege Access

The IAM user docproc-app is used to demonstrate application access. The policy allows listing the documents/ prefix and reading or uploading objects within that prefix in the primary bucket docuvault-prod.

Figure 3.1 — IAM Policy



3.2 S3 Bucket Policy and Access Point

The S3 bucket policy and the access-point policy provide resource-level access control. DocuVault uses the S3 Access Point docuvault-processing-ap to provide a dedicated access path for document-processing operations.

Access is restricted to the permissions explicitly granted by the applicable IAM and resource policies. Public access is blocked on the primary document bucket.

Figure 3.2 — S3 Access Point Policy

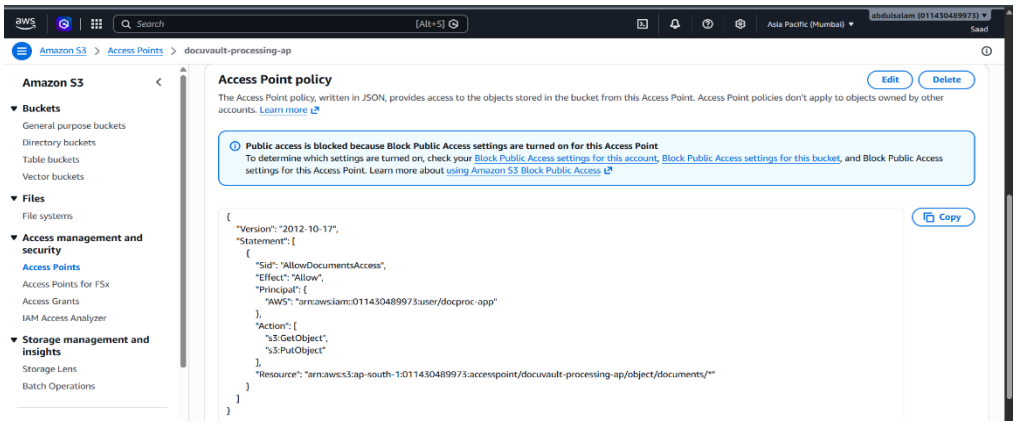
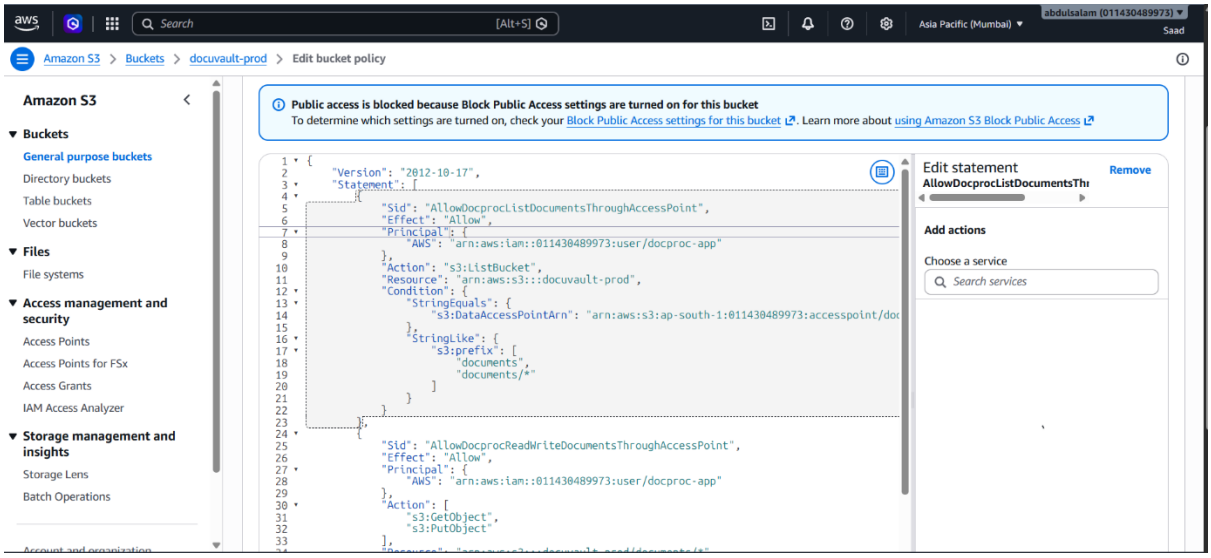


Figure 3.3 — S3 Bucket Policy

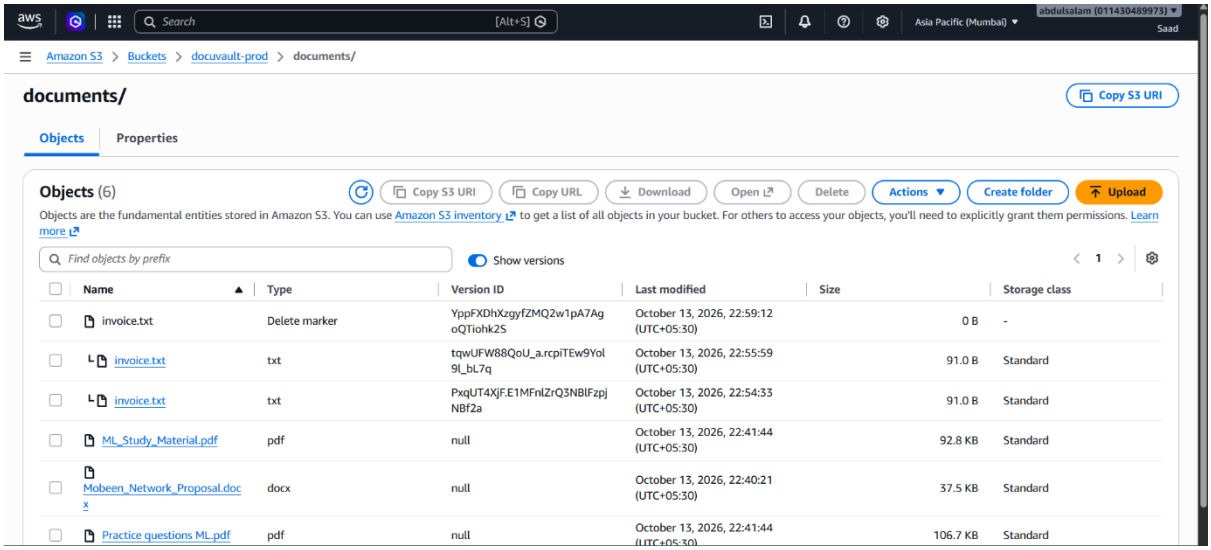


4. Evidence of Version Recovery, Pre-signed URL Access, and Lifecycle Configuration

4.1 S3 Version Recovery

Amazon S3 Versioning was enabled on the primary bucket docuvault-prod. Versioning preserves multiple versions of an object, allowing an earlier version to be recovered after accidental deletion or modification.

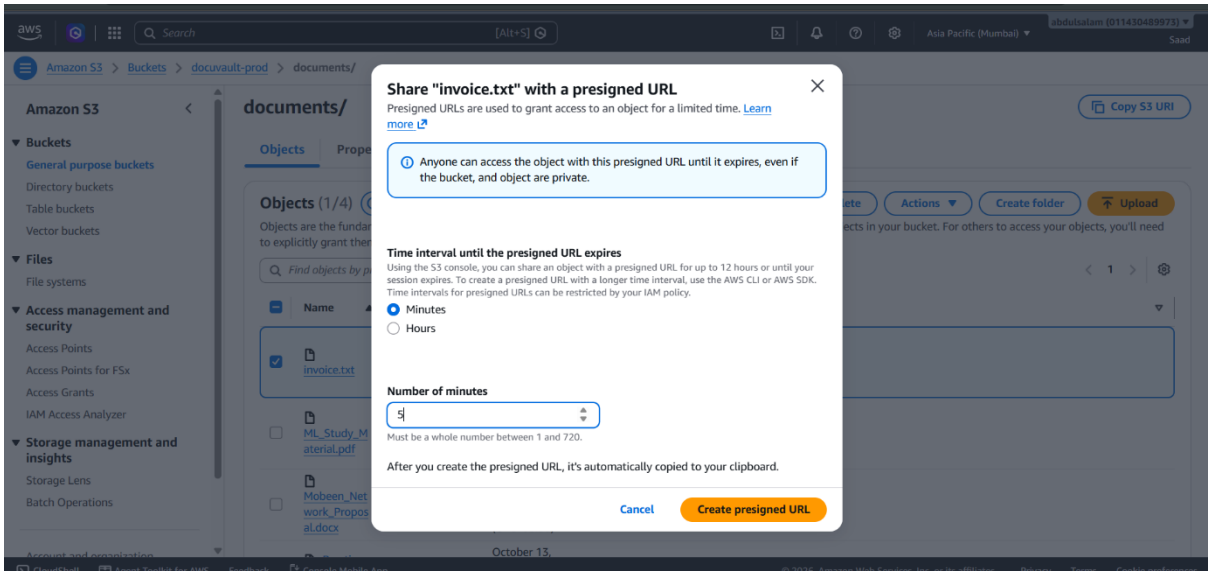
A recovery test was performed using the object documents/version-recovery.txt. After a deletion test, the original file was retrieved successfully, and its original content was verified.



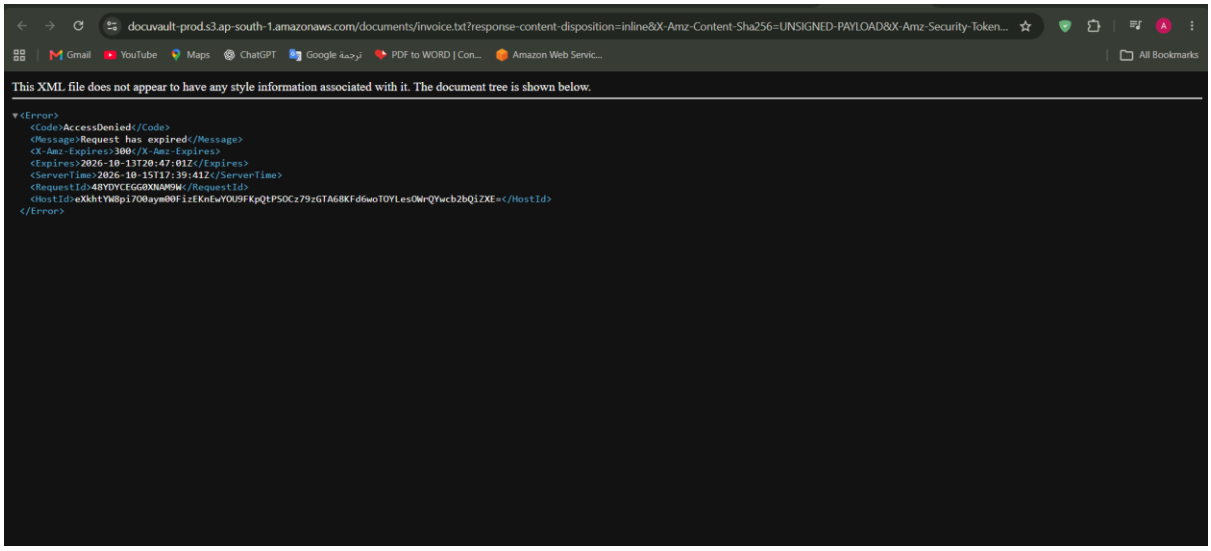
4.2 Pre-signed URL Access

Amazon S3 pre-signed URLs provide temporary access to private objects without requiring the recipient to have direct AWS credentials. The URL is valid for a specified period.

A pre-signed URL was generated for an object in the docuvault-prod bucket with an expiry time of five minutes. The URL was used to test temporary access to the object.





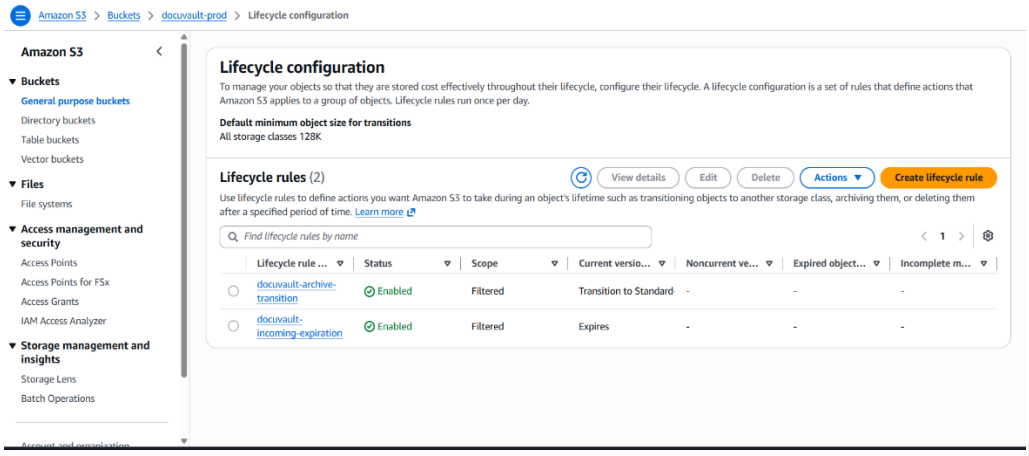


The DocuVault test page returned HTTP status 200 and displayed the invoice content. This demonstrates a successful request. The test was performed using an S3 pre-signed URL, which provides temporary access to a private object.

4.3 S3 Lifecycle Configuration

Amazon S3 Lifecycle Management helps control storage costs by automatically transitioning eligible objects to lower-cost storage classes or expiring objects according to configured rules.

The lifecycle configuration of docuvault-prod was reviewed in the Amazon S3 Console under **Management → Lifecycle rules**.



5. Cross-Region Replication Evidence

5.1 Replication Configuration

Amazon S3 Cross-Region Replication (CRR) is configured to maintain a replica of the primary DocuVault bucket in a separate AWS Region. This improves data protection and supports disaster recovery if the primary Region becomes unavailable.

The replication configuration is as follows:

Parameter	Configuration
Source bucket	docuvault-prod
Source Region	Mumbai (ap-south-1)
Destination bucket	docuvault-dr-011430489973-ap-south-2-an
Destination Region	Hyderabad (ap-south-2)
Replication rule	docuvault-cross-region-replication
Replication status	Enabled
Replication scope	Entire bucket
Replication mode	Asynchronous

5.2 Replication Verification

The replication rule was verified in the Amazon S3 Console and showed an Enabled status. A disaster recovery test was also performed by retrieving the object dr-test.txt from the Hyderabad destination bucket.

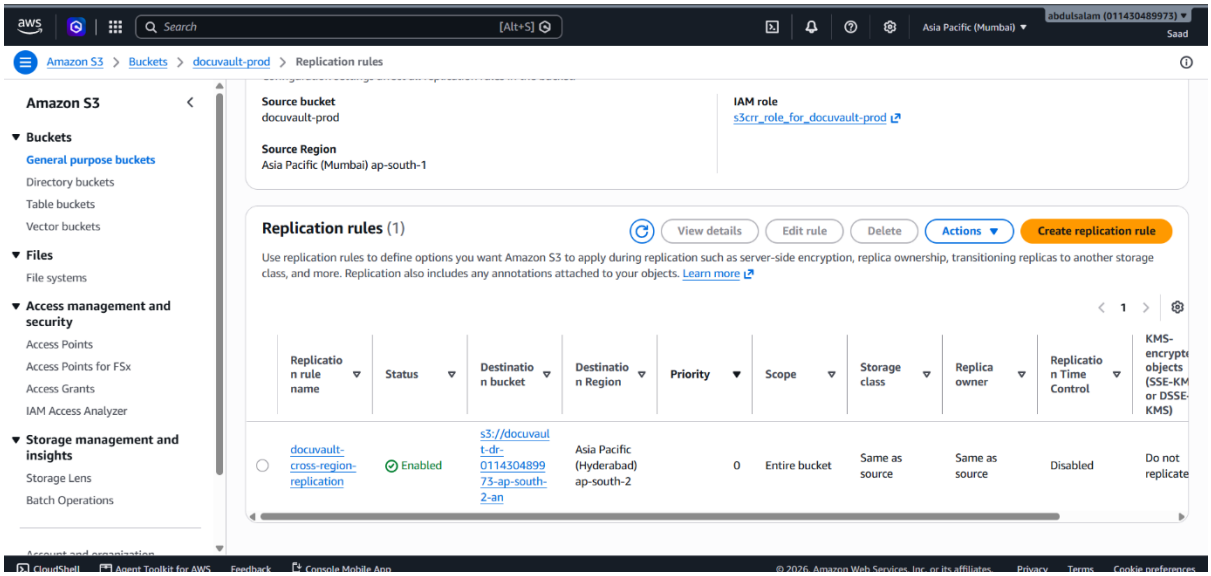
The retrieved object's metadata showed ReplicationStatus: REPLICA, confirming that the object was a replica in the destination bucket.

5.3 Benefits of Cross-Region Replication

- **Disaster recovery:** Provides a separate copy of objects in another AWS Region.

- **Regional resilience:** Helps protect against disruption affecting the primary Region.
- **Data availability:** Allows replicated objects to be retrieved from the destination bucket, subject to permissions and availability.
- **Data protection:** Supports recovery planning and business continuity.

Figure 5.1 — Replication Rule Configuration



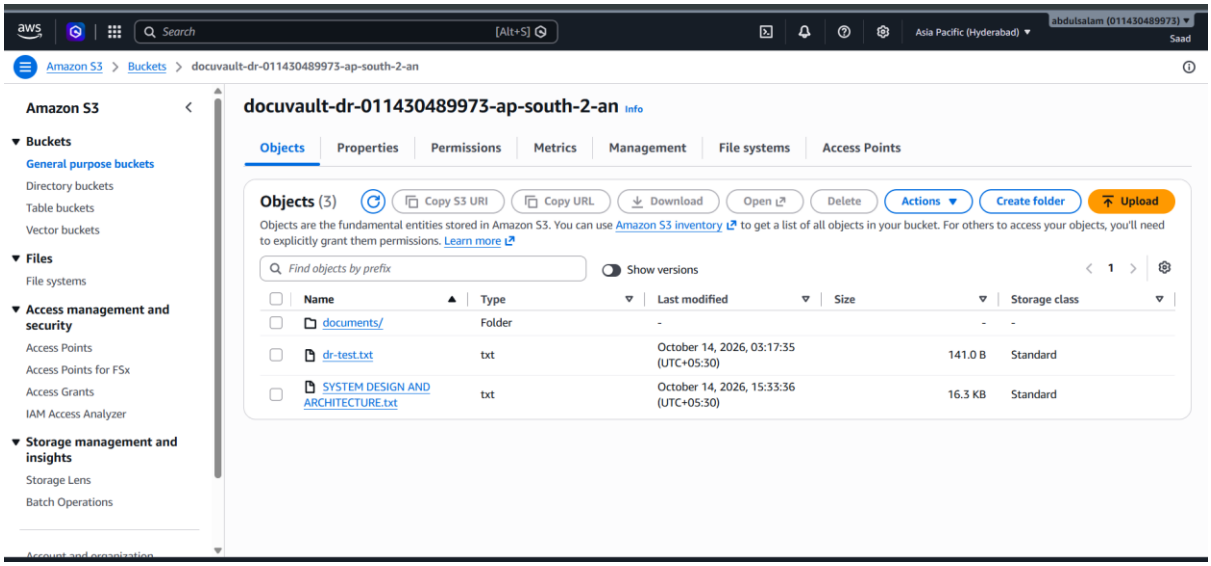
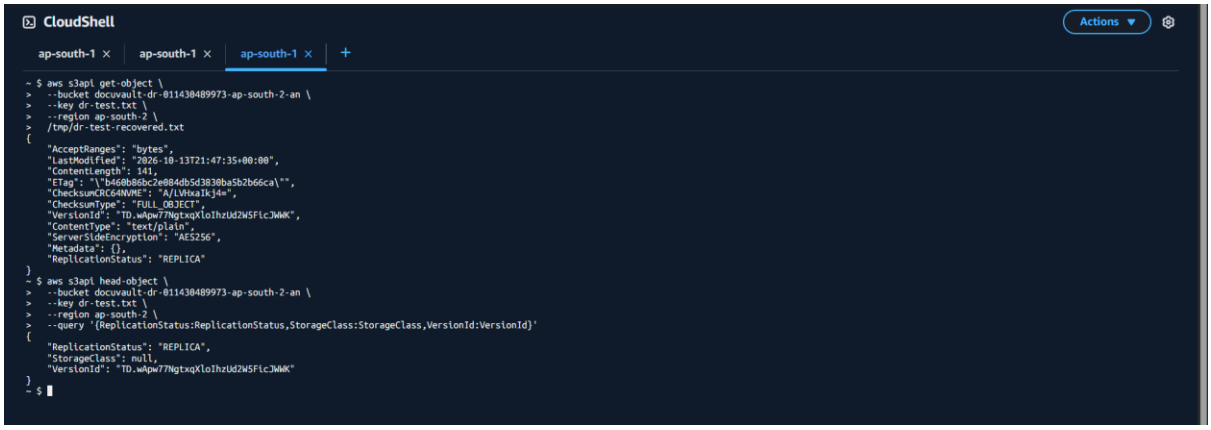


Figure 5.2 — Replicated Object Verification



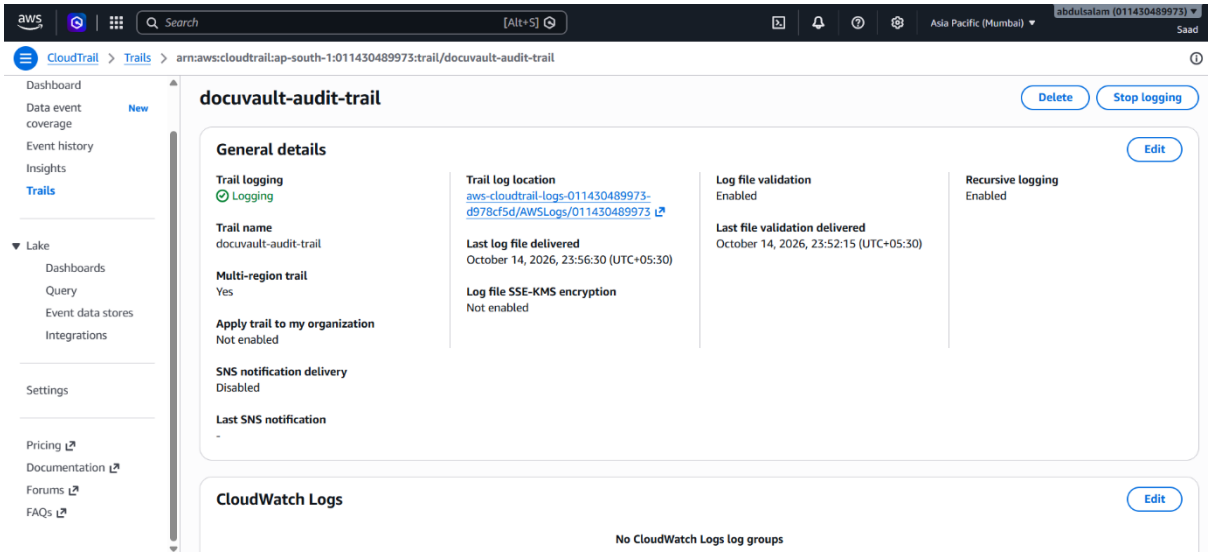
6. Logging and Monitoring Evidence

6.1 AWS CloudTrail Logging

AWS CloudTrail is used to record and audit activities in the DocuVault AWS environment. The trail docuvault-audit-trail is configured and logging is active. Log file validation is enabled, and CloudTrail log files are delivered to a dedicated S3 logging bucket.

CloudTrail data events are configured for object-level activity in the primary bucket, docuvault-prod, supporting investigation of document-related operations.

Figure 6.1 — AWS CloudTrail Logging Configuration.

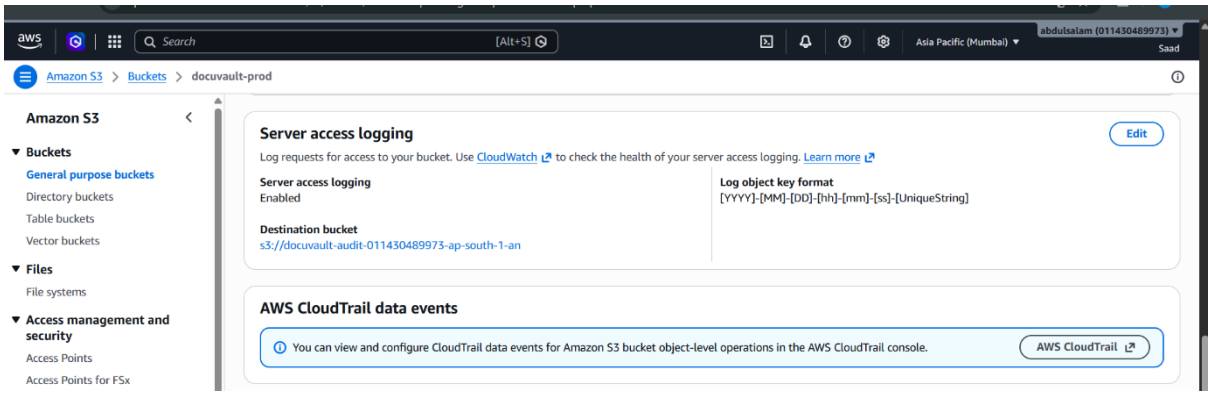


6.2 S3 Server Access Logging

Amazon S3 server access logging is enabled on the primary bucket, docuvault-prod. Access logs are delivered to the dedicated audit bucket, docuvault-audit-011430489973-ap-south-1-an.

These logs help track requests made to the bucket and support security auditing and troubleshooting.

Figure 6.2 — S3 Server Access Logging Configuration.



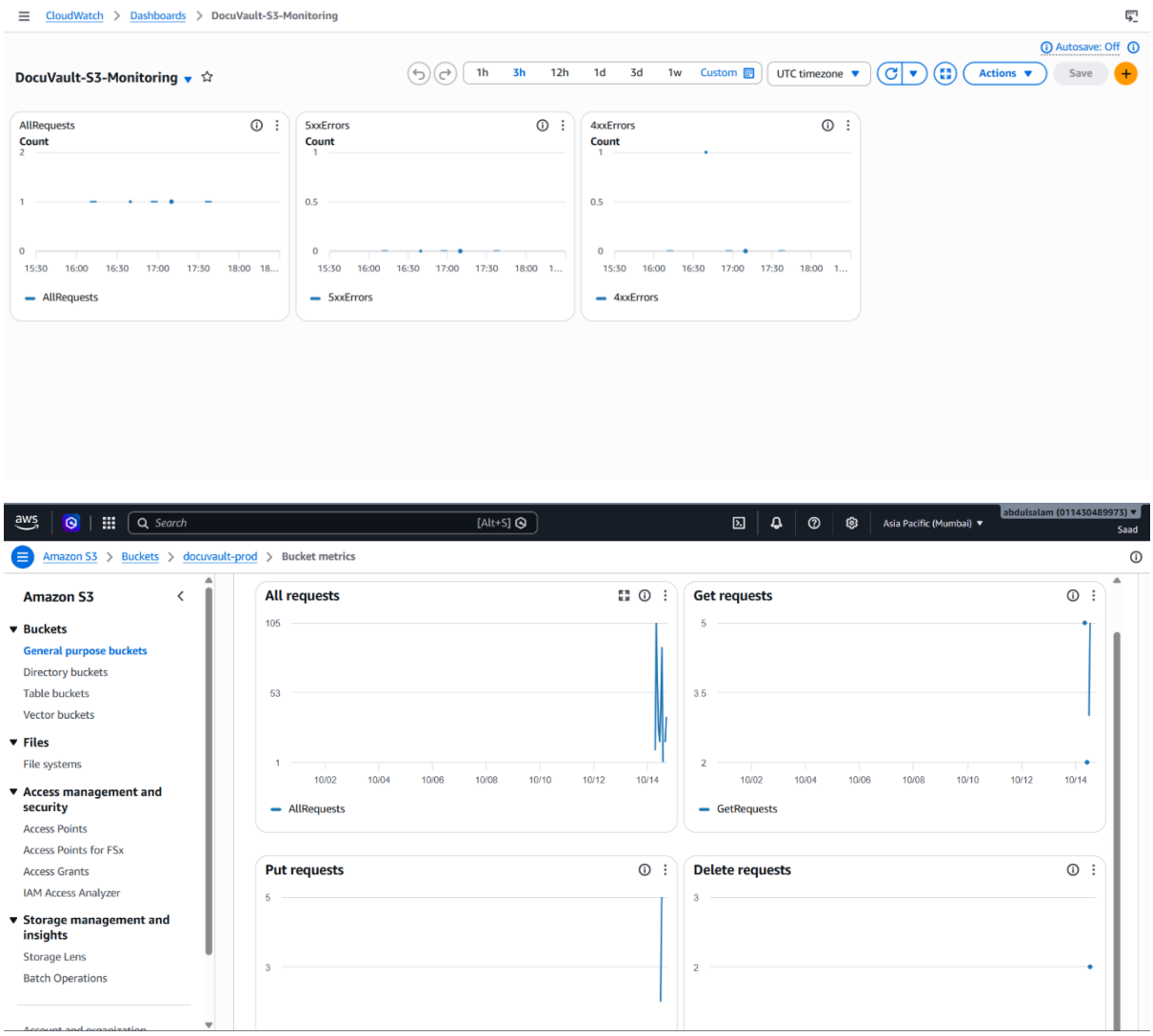
6.3 Amazon CloudWatch Monitoring Dashboard

Amazon CloudWatch is used to monitor S3 request activity. The dashboard DocuVault-S3-Monitoring contains graphs for:

- AllRequests — overall request activity.
- 4xxErrors — client-side request errors.
- 5xxErrors — server-side request errors.

The dashboard provides visibility into request activity and helps identify potential problems.

Figure 6.3 — DocuVault CloudWatch Monitoring Dashboard.

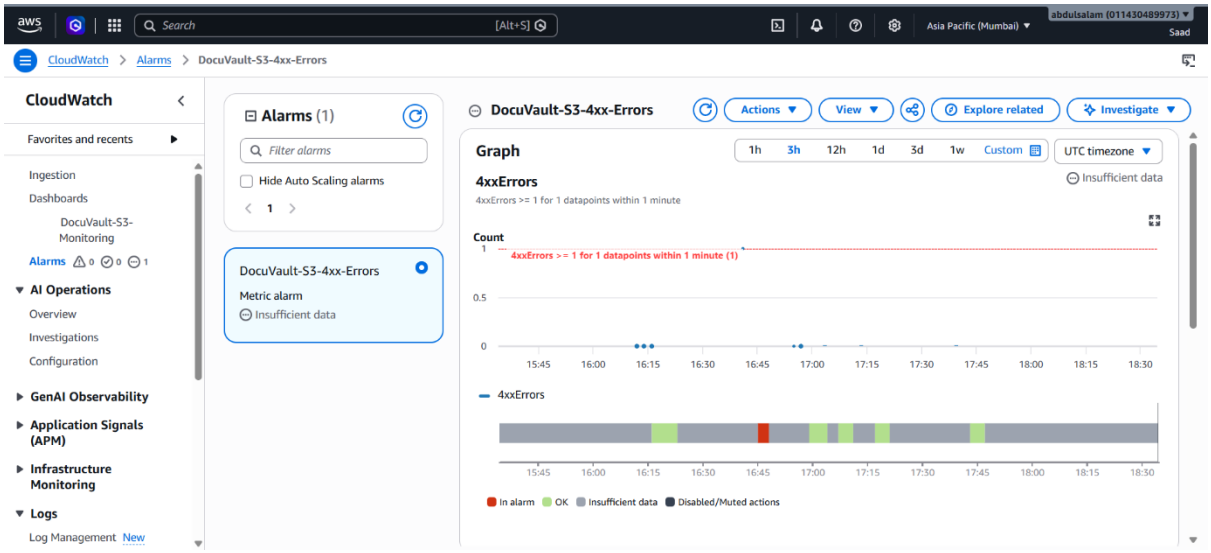


6.4 CloudWatch Alarm Configuration

A CloudWatch metric alarm named DocuVault-S3-4xx-Errors is configured to monitor the 4xxErrors metric. Its configured threshold is at least one error datapoint within one minute.

During evidence collection, the alarm displayed **Insufficient data**. This indicates that CloudWatch did not have enough recent metric data to determine the alarm state at that time. The alarm configuration exists, but its operational state should be verified with additional metric data.

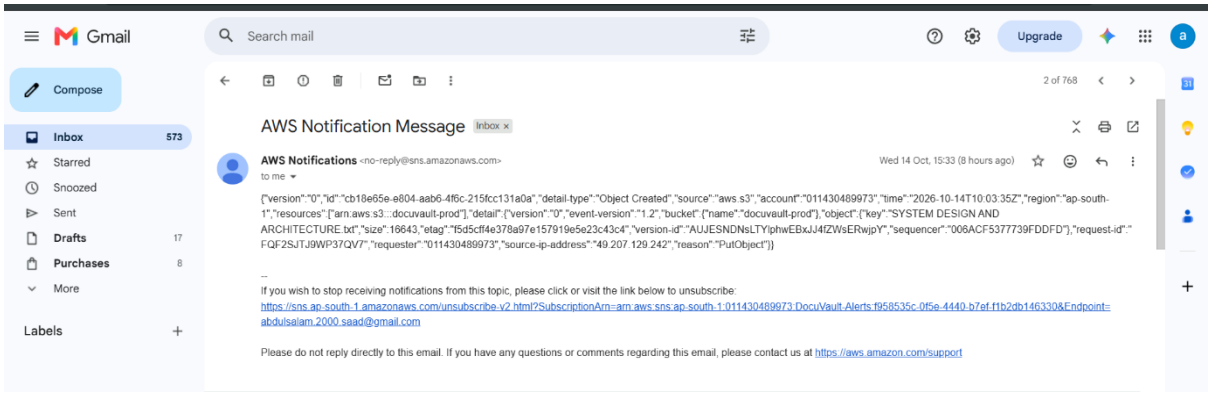
Figure 6.4 — CloudWatch S3 Error Alarm Configuration.



6.5 S3 Event Notification Evidence

An AWS notification email was received for an S3 Object Created event involving the docuvault-prod bucket. This confirms that an object-creation event notification was delivered successfully.

Figure 6.5 — Successful S3 Object-Created Event Notification.



7. Backup and Disaster Recovery Runbook

7.1 Backup Strategy

DocuVault uses Amazon S3 Versioning to preserve previous object versions and Cross-Region Replication (CRR) to maintain copies of objects in a separate AWS Region.

The primary bucket is docuvault-prod in Mumbai (ap-south-1), and the disaster recovery bucket is docuvault-dr-011430489973-ap-south-2-an in Hyderabad (ap-south-2).

7.2 Recovery Point Objective (RPO) and Recovery Time Objective (RTO)

The project proposes the following recovery targets:

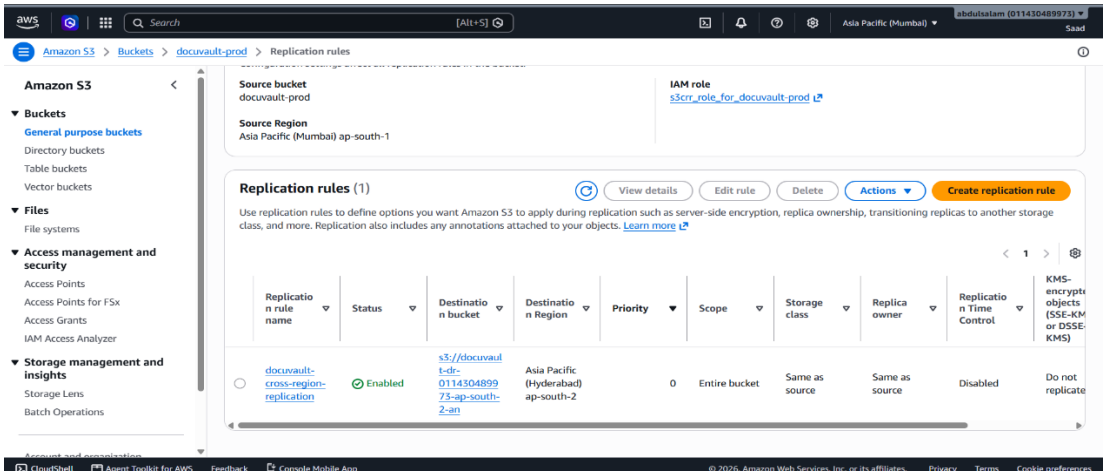
- **RPO: 15 minutes** — the target maximum amount of recent data loss during a disruption.
- **RTO: 4 hours** — the target time to restore essential document access.

These are proposed targets, not measured or guaranteed recovery times. Actual results depend on replication delays, the scope of the incident, permissions, and recovery procedures.

7.3 Recovery Test Evidence

The object dr-test.txt was retrieved successfully from the Hyderabad DR bucket. Its metadata showed ReplicationStatus: REPLICA, confirming that a replicated object was available for retrieval.

Figure 7.1 — Cross-Region Replication Rule



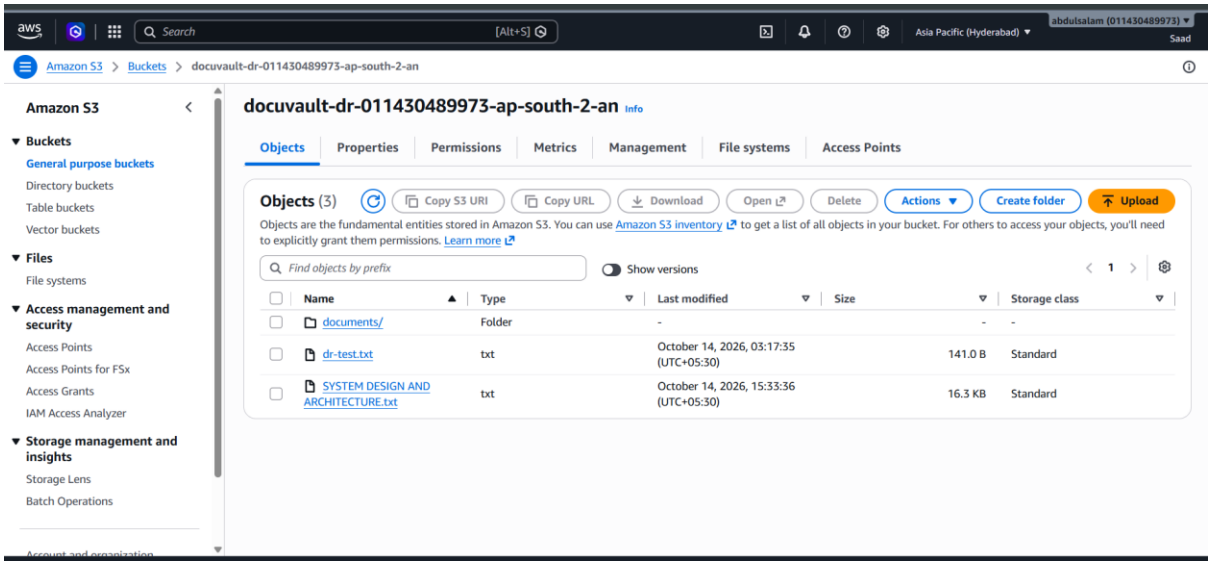
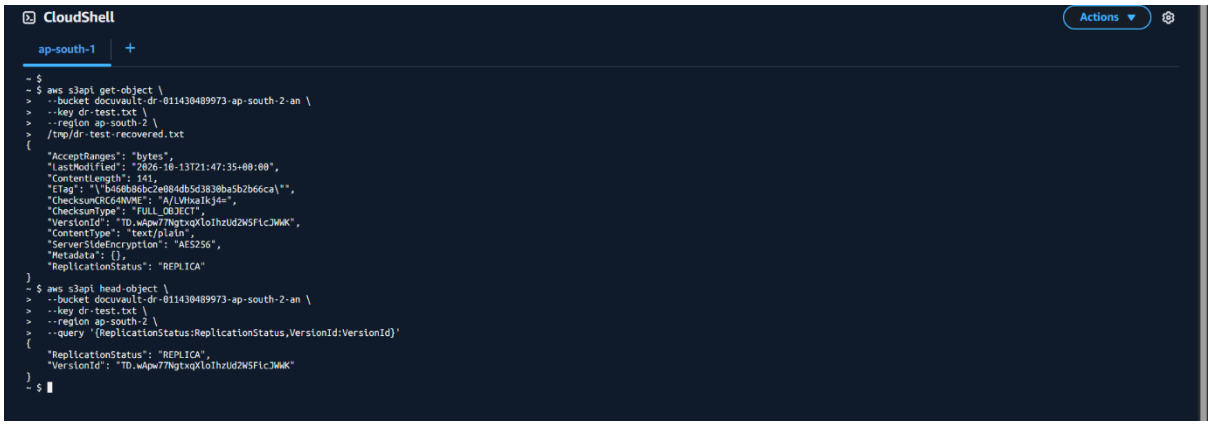


Figure 7.2 — Disaster Recovery Object Retrieval Verification



8: Incident Report

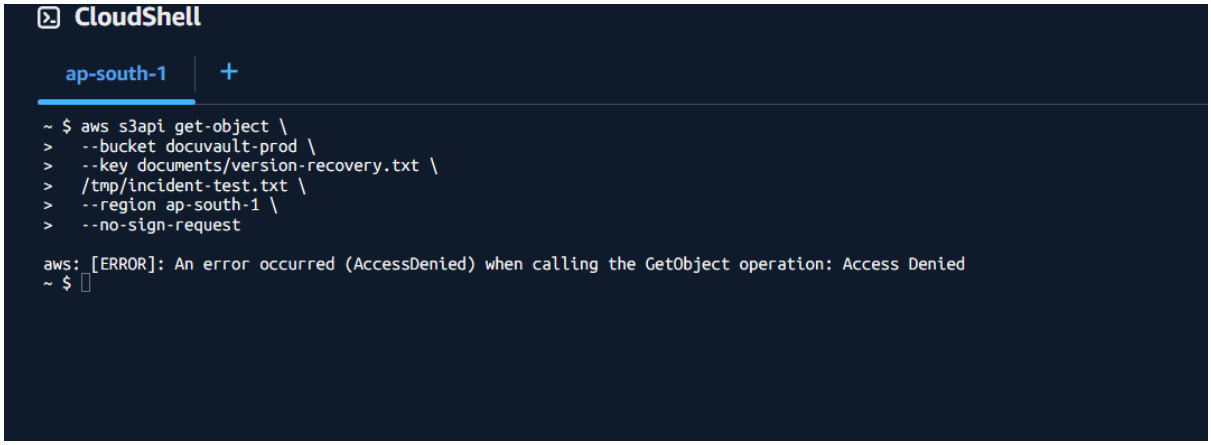
8.1 Incident Description

The incident simulated in this project was an unauthorized access attempt to a private document stored in Amazon S3. The affected bucket was docuvault-prod in the Mumbai region (ap-south-1).

8.2 Incident Simulation and Result

An unsigned request was made to download documents/version-recovery.txt without AWS authentication. The request returned an AccessDenied error. This demonstrated that the document could not be retrieved through the unauthenticated request.

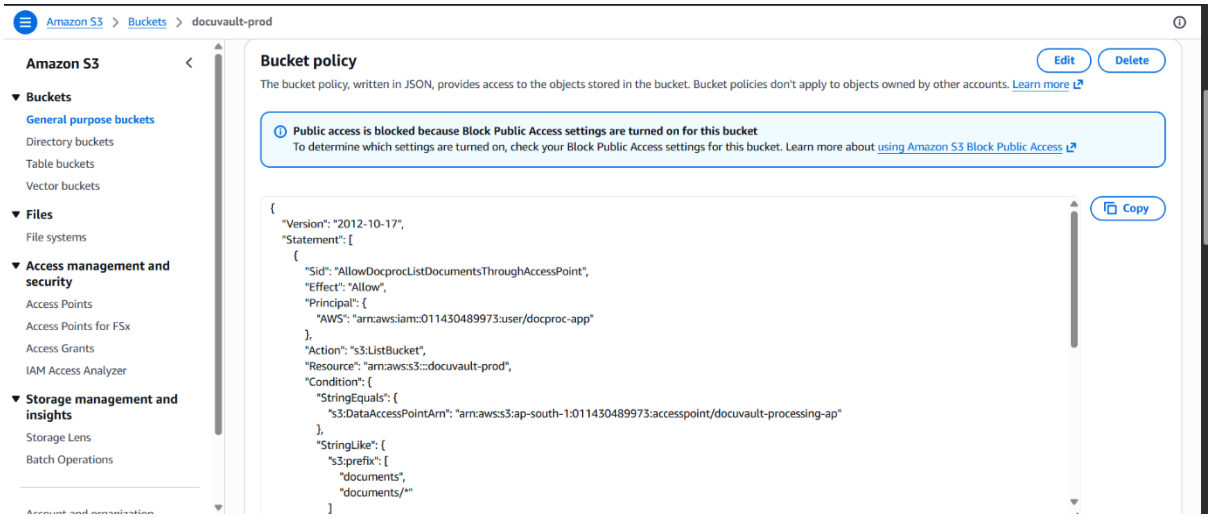
Figure 8.1 — Unauthorized Access Attempt



8.3 Security Controls

The S3 bucket's Block Public Access settings were reviewed to verify that public access was blocked. Access to private documents is intended to be controlled through IAM permissions and the approved S3 Access Point.

Figure 8.2 — S3 Block Public Access Configuration



Deliverable 9: Cost-Optimization Proposal

9.1 Objective

The objective of cost optimization is to reduce Amazon S3 storage expenses while maintaining data security, availability, and disaster recovery for the DocuVault document storage system.

9.2 Storage Class Strategy

Different Amazon S3 storage classes can be used depending on how frequently documents are accessed.

Storage class	Purpose	Cost-optimization strategy
S3 Standard	Frequently accessed documents	Store active documents here for fast access.
S3 Standard-IA	Infrequently accessed documents	Move older documents here when access is less frequent.
S3 Glacier Flexible Retrieval	Long-term archived documents	Use for documents that can tolerate restoration delays.
S3 Glacier Deep Archive	Very rarely accessed documents	Use for long-term retention when the lowest storage cost is the priority.

9.3 Lifecycle Management

DocuVault uses S3 Lifecycle rules to automate storage management. Lifecycle policies can transition eligible objects to lower-cost storage classes or expire temporary files after the required retention period.

The primary bucket, docuvault-prod, has two enabled lifecycle rules:

- docuvault-archive-transition
- docuvault-incoming-expiration

These rules support automated storage management. The exact transition and expiration periods should be verified in the AWS Console before being reported.

9.4 Retention and Recovery

S3 Versioning is enabled on the primary bucket to support recovery from accidental overwrites or deletions. Noncurrent versions should be retained according to business and recovery requirements, with lifecycle rules used to manage their storage costs where appropriate.

Important documents should not be expired until their retention requirements have been checked. The disaster recovery bucket in Hyderabad provides an additional copy through Cross-Region Replication.

9.5 Additional Cost-Control Measures

1. Review S3 storage usage and request metrics regularly.
2. Remove incomplete multipart uploads through an appropriate lifecycle rule.
3. Apply expiration policies to temporary files only after their required retention period.
4. Use S3 Storage Lens or S3 metrics to identify unnecessary storage growth.
5. Review replication, logging, retrieval, and data-transfer charges.
6. Keep only the versions and archived records required by the project's recovery and retention policies.

9.6 Conclusion

DocuVault can optimize costs by matching storage classes to access frequency, automating lifecycle management, and applying appropriate retention policies. These measures help control storage growth without compromising document security or disaster recovery.

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